

PINOCCHIO-hodpy: frozen-model diagnostic audit

Real L3870N4096/000 lightcone | 660,873 galaxies | 5,435,239 native halo occurrences
No HOD, colour, satellite, or cosmological parameter was changed. No fitting was performed.

Reference classes: only the SDSS/GAMA luminosity function is an independently specified observational target. HOD occupations, colours, NFW radii and Gaussian velocities are predictions of the model under test. The native HMF is measured simulation input. Clustering is an unfitted diagnostic, not a validation against observations.

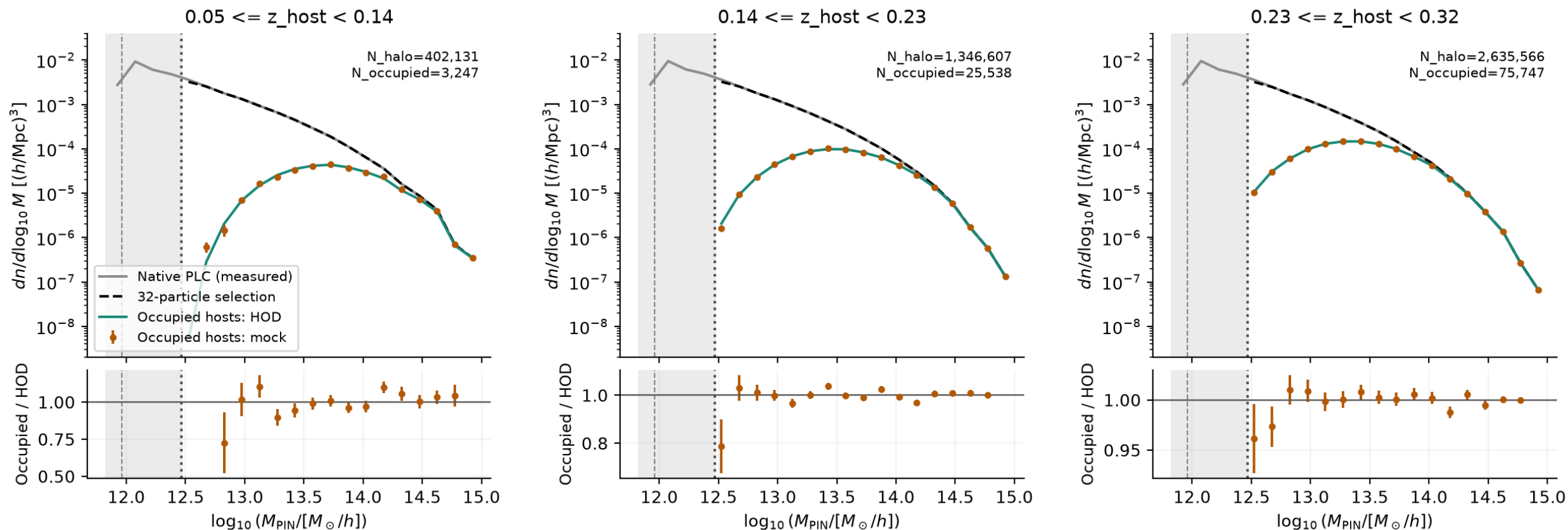
Conventions: M_PIN is native fragmentation-group mass [Msun/h], not a spherical-overdensity mass. Native floor: 10 particles; painting cut: 32 particles = 2.93126e12 Msun/h. Positions: comoving Mpc/h; velocities: proper peculiar km/s. AB magnitudes: $^0.1 M_r - 5\log_{10}(h)$; rest-frame colour: $^0.1(g-r)$. Inner cone: 65 deg. LF claims: $M_r \leq -22$ and $0.05 \leq z_{\cos} < 0.32$; $r < 20$ where labelled. The underlying $M_r < -21.5$ catalogue is not a complete faint flux-limited survey.

Cosmology: $\Omega_m=0.3913$, $\Omega_b=0.0419$, $\Omega_{DE}=0.6087$, $h=0.7276$, $n_s=0.9312$, $\sigma_8=0.60850325$, $w_0=-1.168$, $w_a=-0.6605$. Simulation seed=1386; galaxy seed=20260917.

z range	M_r interval	N	HOD residual	Mock residual	Allowed	Status
0.05-0.14	[-22.8,-22.5)	246	+0.0060%	+0.485%	23.26%	PASS
0.05-0.14	[-22.5,-22.2)	1082	+0.0043%	+0.816%	10.60%	PASS
0.05-0.14	[-22.2,-22.0)	1904	+0.0032%	-3.026%	9.53%	PASS
0.14-0.23	[-22.8,-22.5)	2237	+0.0053%	+1.023%	8.17%	PASS
0.14-0.23	[-22.5,-22.2)	8947	+0.0034%	+0.320%	5.47%	PASS
0.14-0.23	[-22.2,-22.0)	14665	+0.0022%	-0.124%	5.00%	PASS
0.23-0.32	[-22.8,-22.5)	7214	+0.0048%	-0.196%	5.00%	PASS
0.23-0.32	[-22.5,-22.2)	27260	+0.0029%	+0.510%	5.00%	PASS
0.23-0.32	[-22.2,-22.0)	42565	+0.0016%	-0.052%	5.00%	PASS

LF uncertainty: max(conditional HOD sigma, 32-region spatial-jackknife sigma); tolerance max(5%,3 sigma). Full densities, volumes, errors and completeness estimates are saved in CSV files. Every following plot states its own selection, sample size, units and uncertainty model.

01 Native HMF, halo selection and occupied hosts



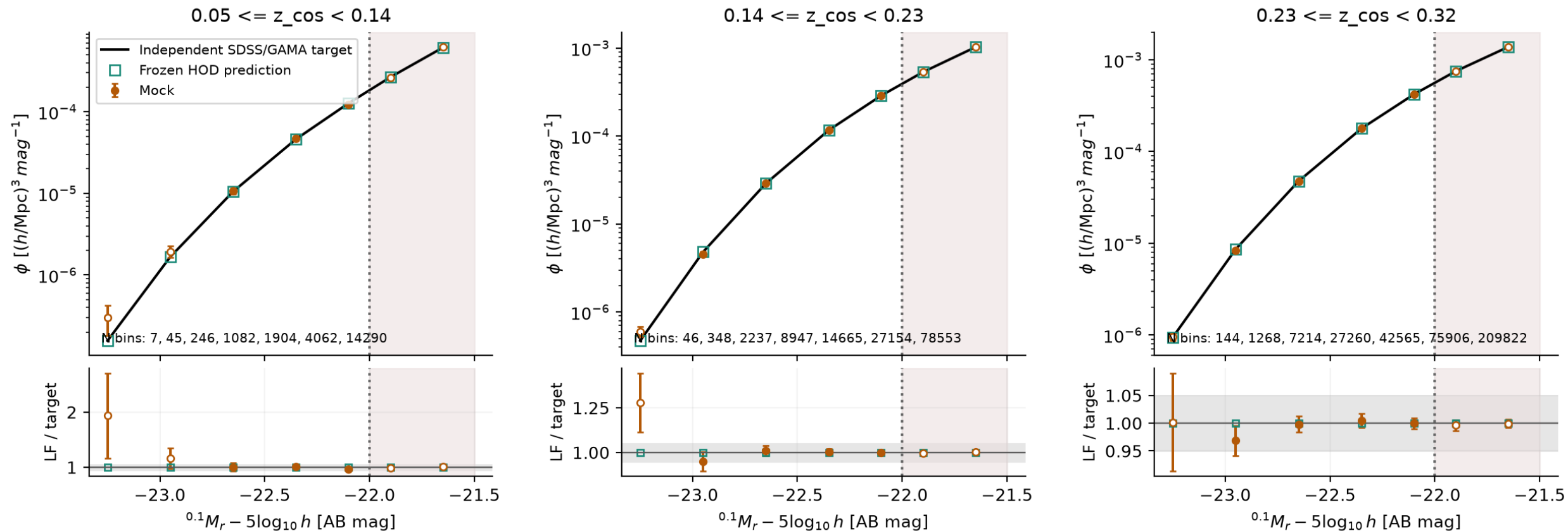
L3870N4096/000: native fragmentation-group M_{PIN} [Msun/h], NOT M_{200m} ; native floor 10 particles; painting cut 32 particles = $2.931e12$ Msun/h. Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.

Selection: inner 65 deg, z_{host} bins; HMF: all native halos; occupied hosts: $M_r < -22$, before $r < 20$ | Sample: $z=0.05-0.14$; $z=0.14-0.23$; $z=0.23-0.32$

Uncertainty: occupied-host Bernoulli 1 sigma; ratios shown only for expected counts ≥ 5 ; grey = below painting cut

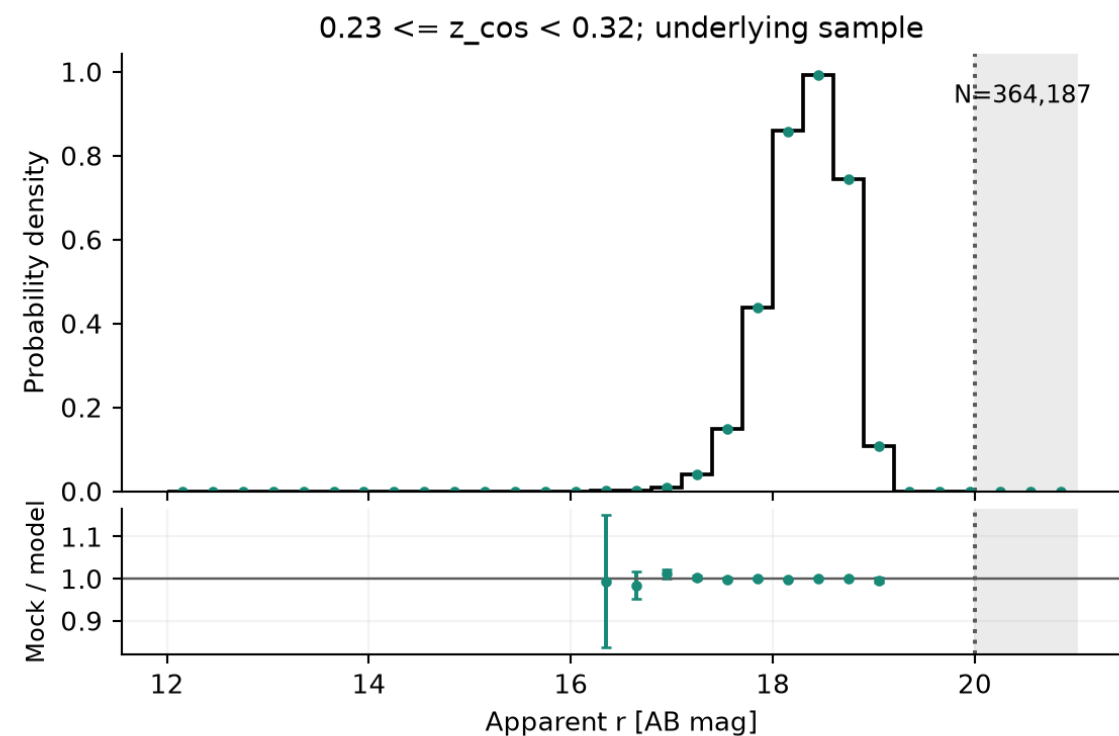
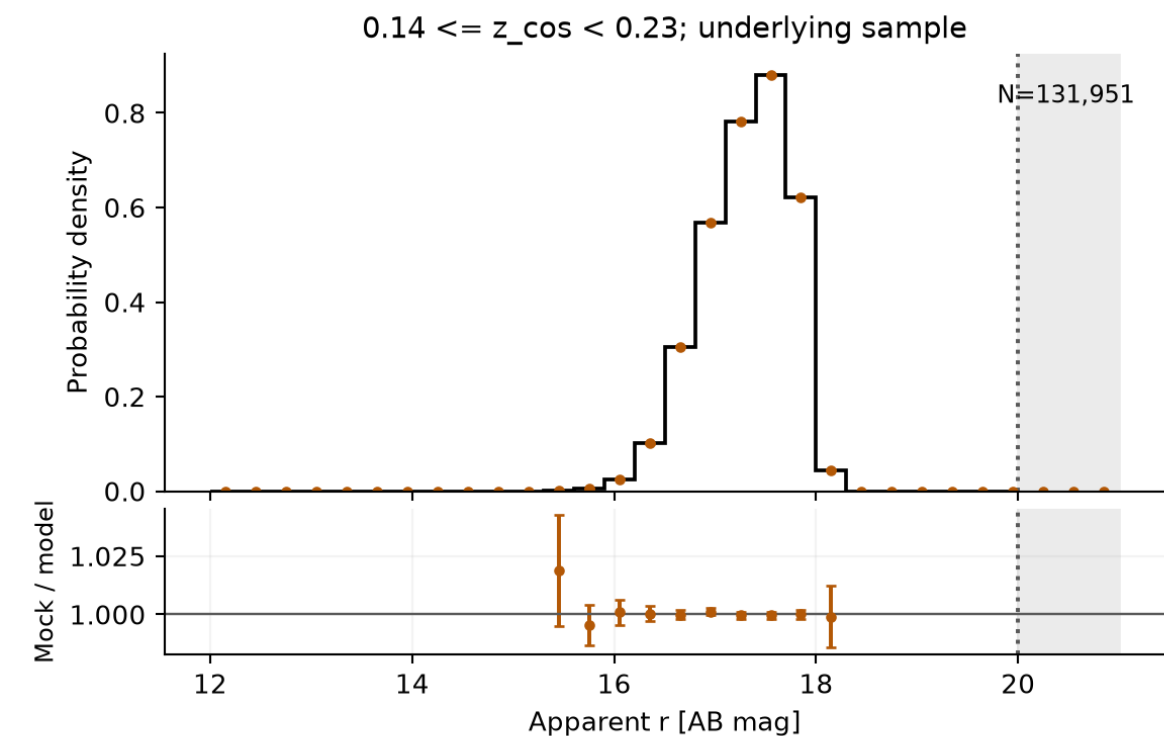
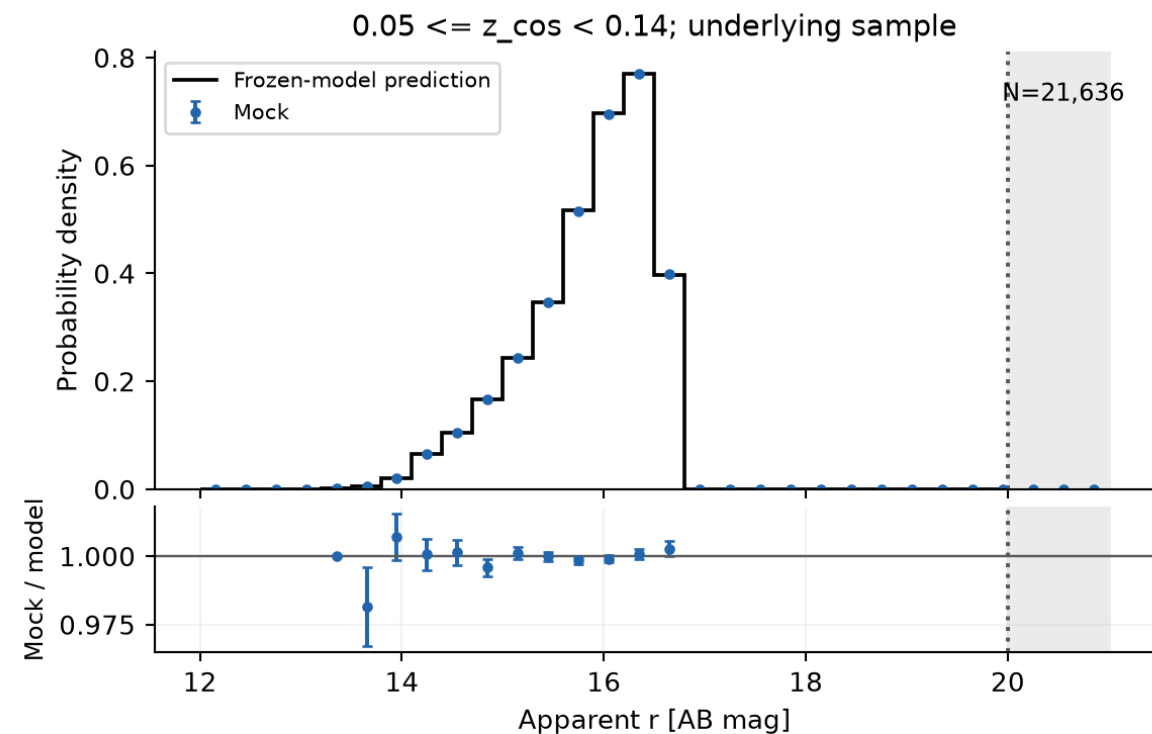
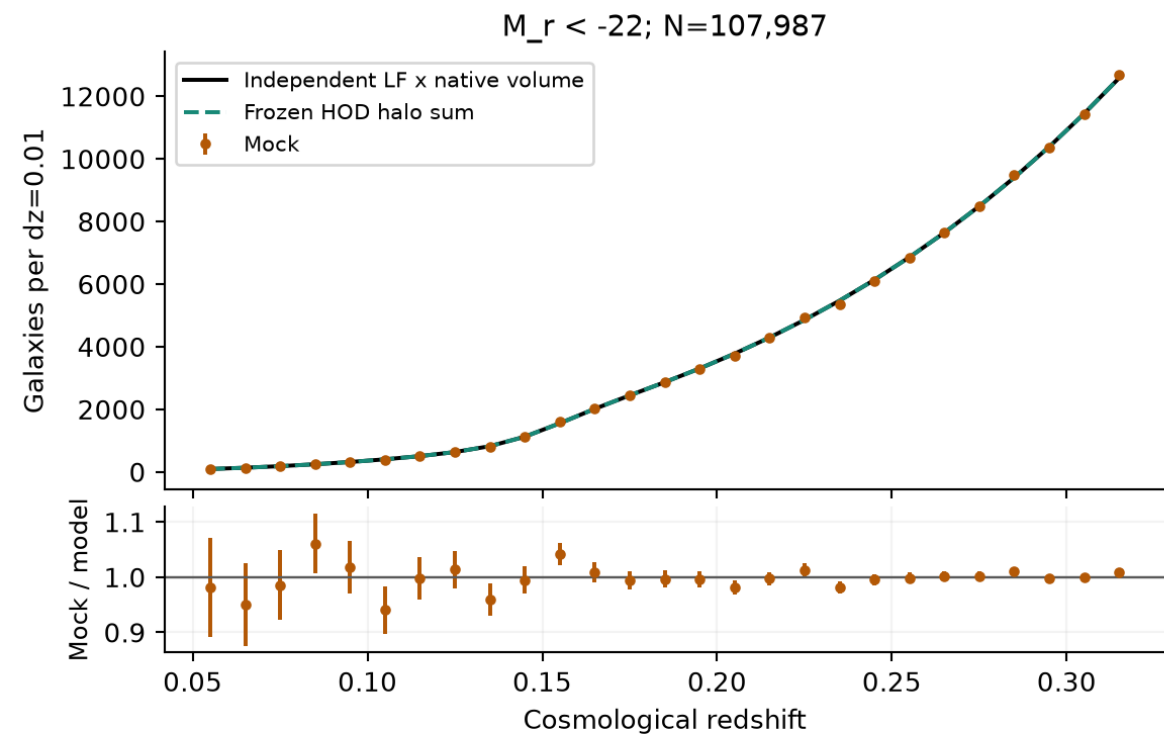
Reference: native HMF is measured input, NOT an independent analytic HMF; occupied-host curve is the tested HOD. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

02 Luminosity-function closure to an independent target



L3870N4096/000: native fragmentation-group M_{PIN} [Msun/h], NOT M_{200m} ; native floor 10 particles; painting cut 32 particles = $2.931e12$ Msun/h.
 Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.
 Selection: inner 65 deg; $r < 20$; z_{cos} and M_r bins shown; red shading = no completeness claim | Sample: counts per bin printed; 3 required complete bins per z slice
 Uncertainty: 1 sigma = max(HOD Bernoulli+Poisson, 32-region jackknife); open symbols: $N < 200$ or $M_r > -22$
 Reference: SDSS/GAMA LF is independently specified; HOD is calibrated to that target. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

03 Redshift counts and the apparent-magnitude selection



L3870N4096/000: native fragmentation-group M_{PIN} [M_{sun}/h], NOT M_{200m} ; native floor 10 particles; painting cut 32 particles = $2.931e12 M_{\text{sun}}/h$.

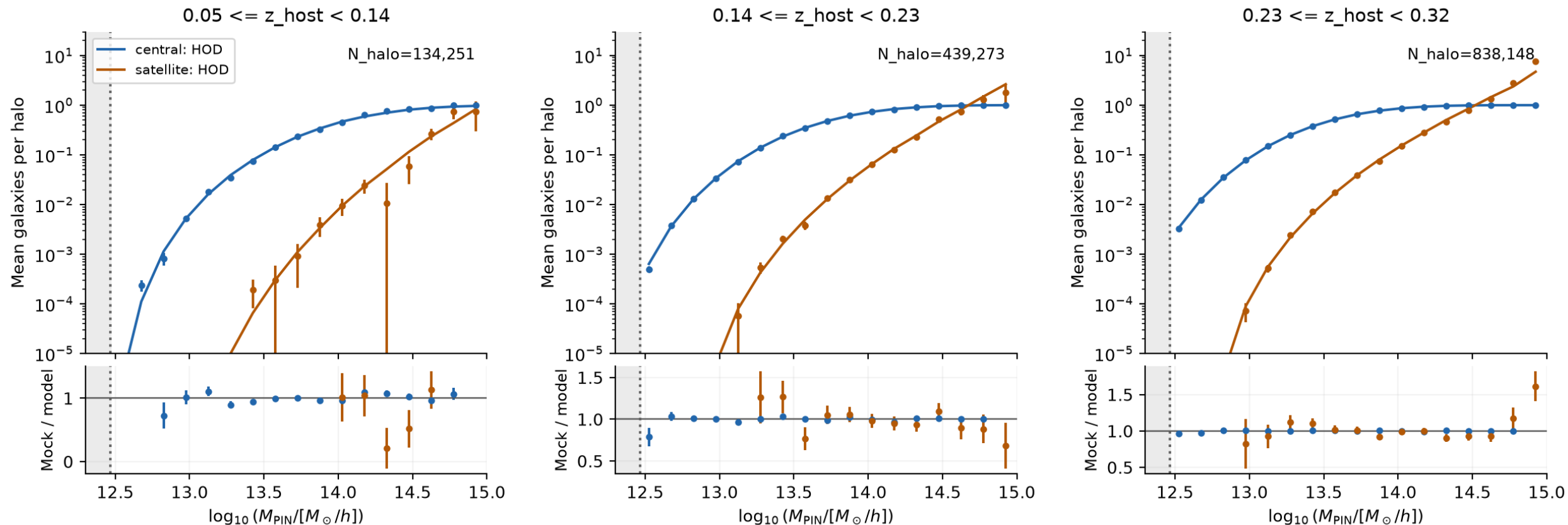
Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.

Selection: inner 65 deg; upper-left $M_r < -22$, $r < 20$; other panels underlying $M_r < -21.5$ BEFORE $r < 20$ | Sample: N printed; $r=20$ line is NOT a claim of faint flux completeness

Uncertainty: upper-left: conditional HOD 1 sigma; apparent- r : conditional colour-draw 1 sigma (N and M_r, z held fixed)

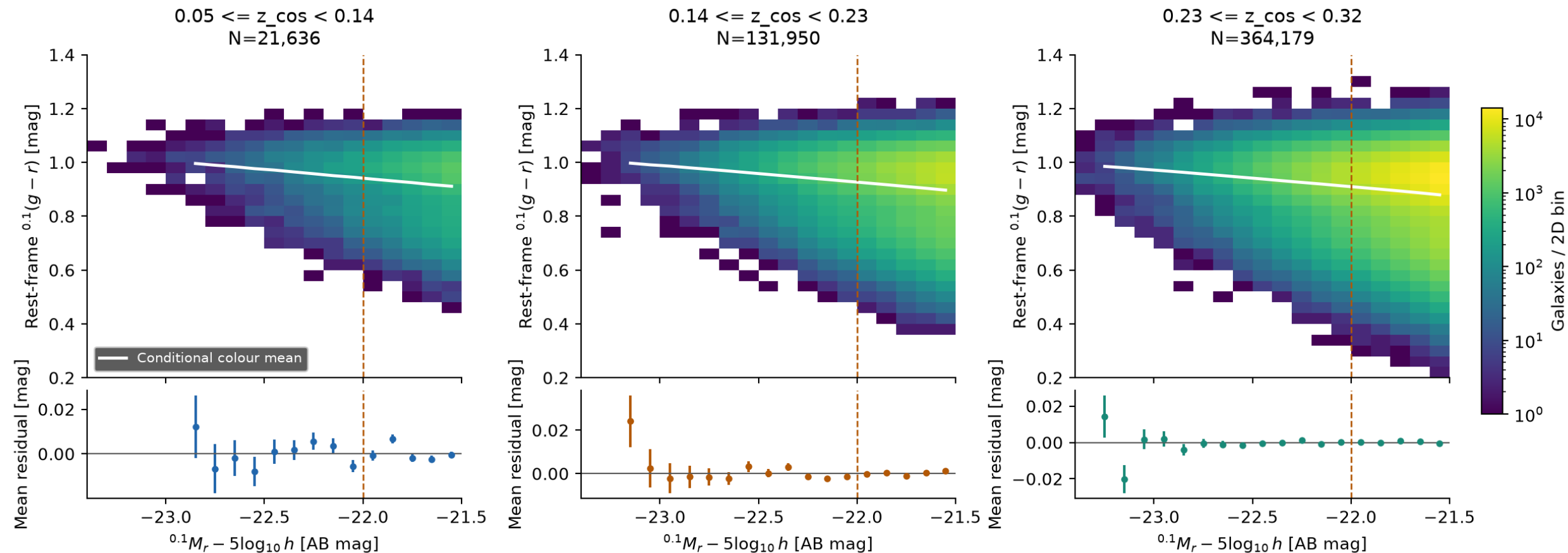
Reference: upper-left independent LF and frozen HOD; other panels analytic colour+k-correction of the same model. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

04 Luminosity-threshold occupations on native halo mass



L3870N4096/000: native fragmentation-group M_{PIN} [Msun/h], NOT M_{200m} ; native floor 10 particles; painting cut 32 particles = $2.931e12$ Msun/h.
 Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.
 Selection: host-selected inner 65 deg and z_{host} ; $M_r < -22$, before $r < 20$; ALL resolved halos in denominators | Sample: $z=0.05-0.14$; $z=0.14-0.23$; $z=0.23-0.32$
 Uncertainty: central Bernoulli and satellite Poisson 1 sigma; lines=HOD, markers=mock; ratios require expectation ≥ 5
 Reference: frozen native-mass HOD averaged over actual masses and redshifts, NOT observed richness. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

05a Rest-frame colour versus luminosity



L3870N4096/000: native fragmentation-group M_{PIN} [Msun/h], NOT M200m; native floor 10 particles; painting cut 32 particles = $2.931e12$ Msun/h.

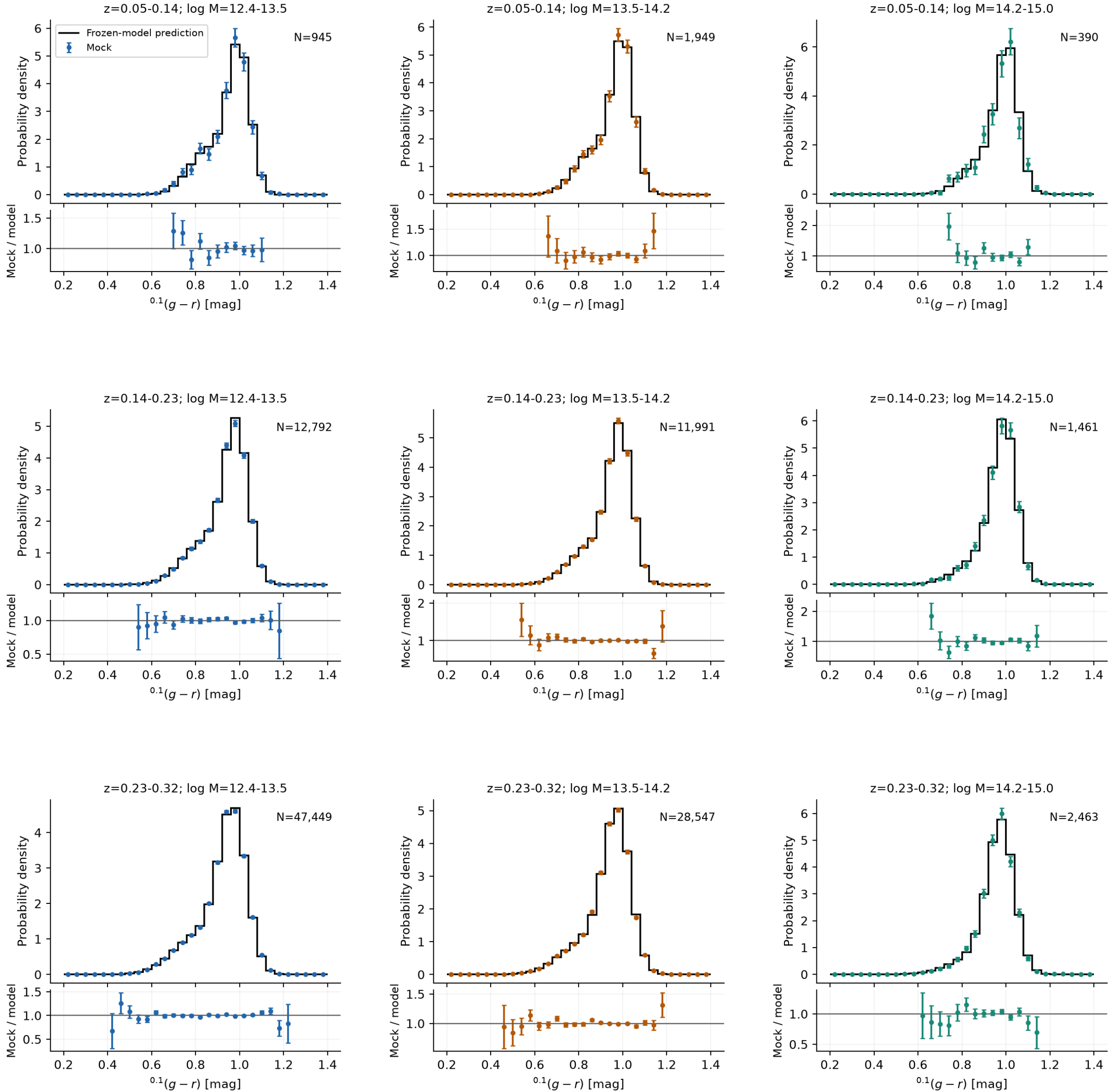
Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.

Selection: inner 65 deg; z_{cos} bins shown; underlying $M_r < -21.5$ BEFORE flux cut; right of dashed line not LF-complete | Sample: N inside displayed magnitude/colour range printed; residual bins $N \geq 20$

Uncertainty: mean residual: exact conditional-mixture 1 sigma; count image uses log colour scale, not an uncertainty

Reference: white curve is hodpy's conditional mixture mean, NOT an independently observed colour target. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

05b Colour mixtures across redshift and host mass



L3870N4096/000: native fragmentation-group M_{PIN} [Msun/h], NOT M200m; native floor 10 particles; painting cut 32 particles = $2.931e12$ Msun/h.

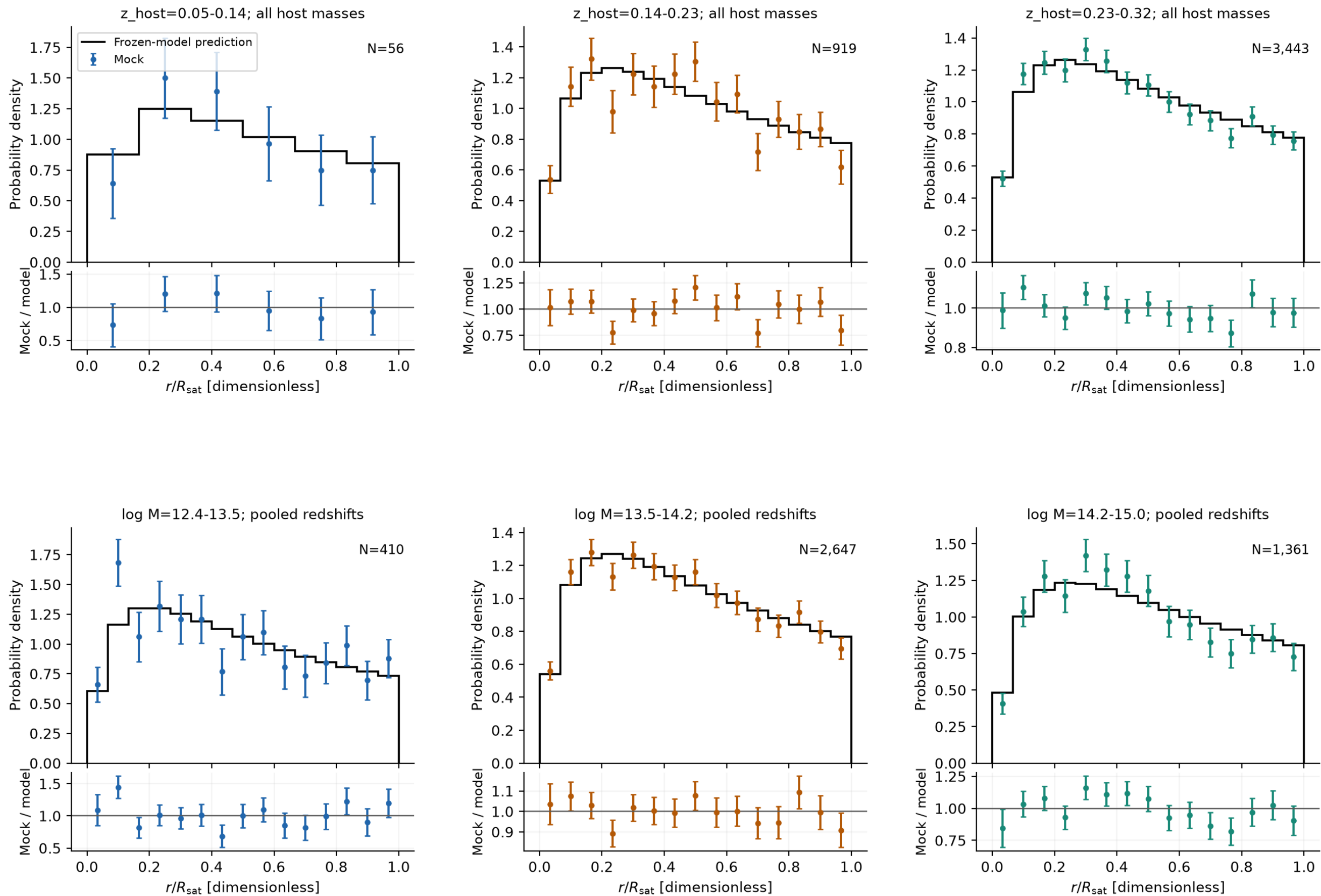
Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.

Selection: inner 65 deg; $M_r < -22$; pre-flux-cut intrinsic sample; z_{cos} and $\log_{10}$ native host mass bins labelled | Sample: N in each panel; 3 redshift x 3 host-mass bins

Uncertainty: conditional Poisson-binomial 1 sigma from Gaussian-mixture bin probabilities; ratios only $E[N] \geq 5$

Reference: integrated upstream hodpy colour mixture at each galaxy's M_r, z and central status; model test only. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

06 Truncated-NFW satellite radii



L3870N4096/000: native fragmentation-group M_{PIN} [Msun/h], NOT M_{200m} ; native floor 10 particles; painting cut 32 particles = $2.931e12$ Msun/h.

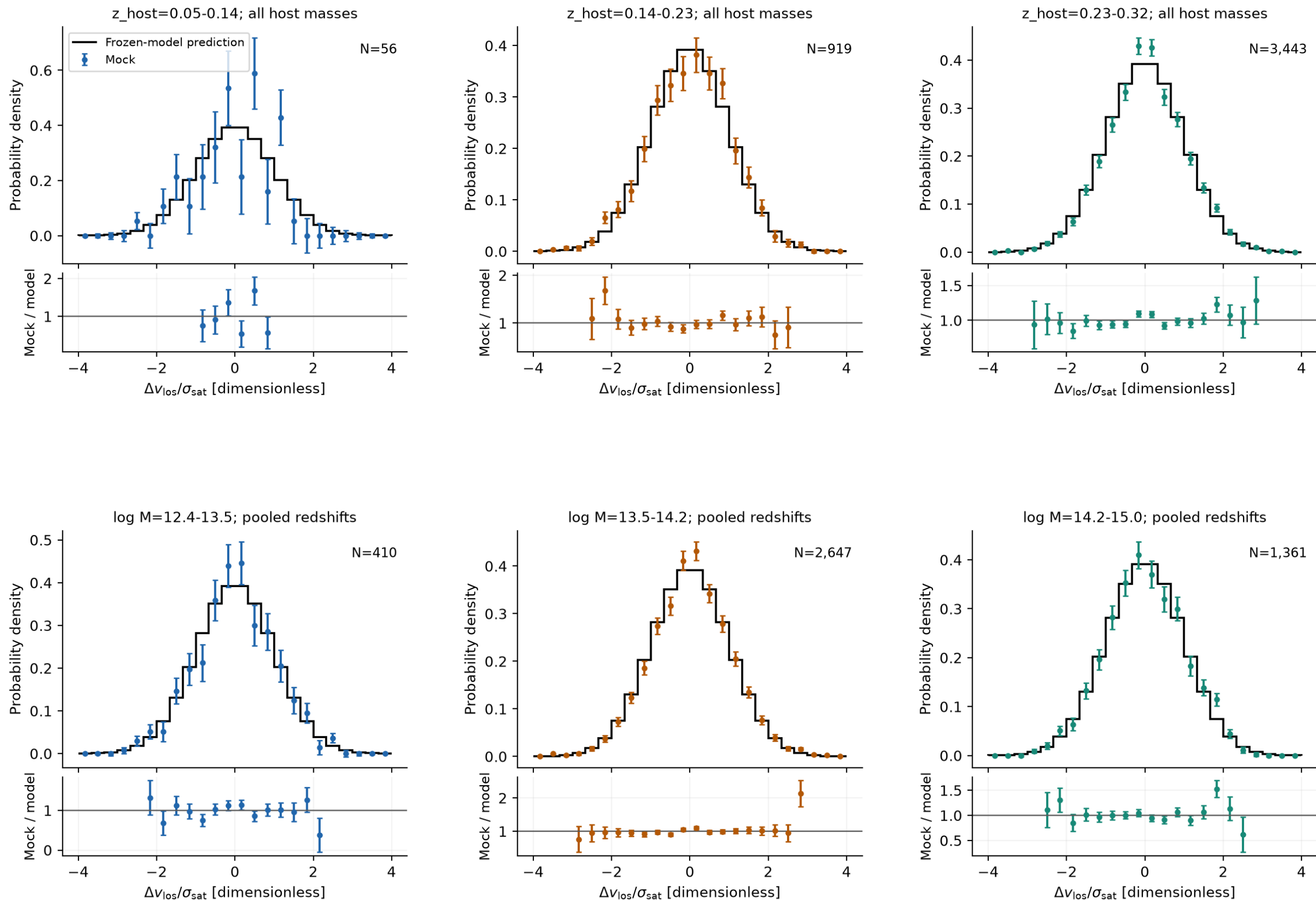
Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.

Selection: SATELLITES; host-selected cone65, $0.05 \leq z_{\text{host}} < 0.32$; $M_r < -22$; pre-flux cut (avoids position/velocity selection) | Sample: N per panel; hard support at $r/R_{\text{sat}}=1$; effective radius is NOT measured R_{200m}

Uncertainty: conditional binomial 1 sigma; ratios require expected bin count ≥ 5 ; low- z satellite sample is small

Reference: exact truncated 3D NFW CDF, averaged over each halo's predicted concentration; internal sampler test. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

07 Satellite velocities and redshift-space prescription



L3870N4096/000: native fragmentation-group M_{PIN} [Msun/h], NOT M200m; native floor 10 particles; painting cut 32 particles = $2.931e12$ Msun/h.

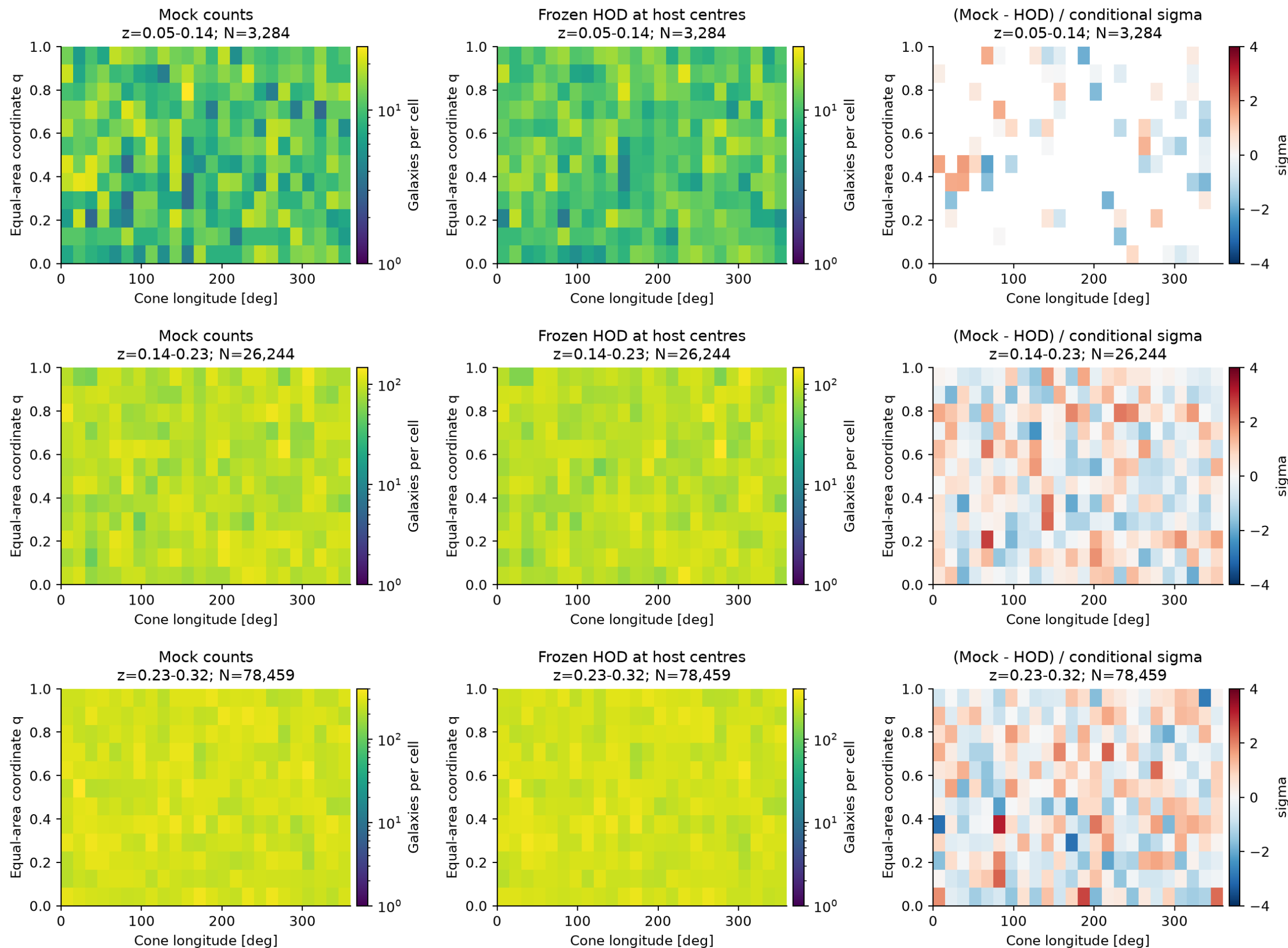
Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.

Selection: SATELLITES; host-selected cone65, $0.05 \leq z_{\text{host}} < 0.32$; $M_r < -22$; pre-flux cut (avoids position/velocity selection) | Sample: N per panel; max Doppler-redshift closure error= $2.95e-17$

Uncertainty: conditional binomial 1 sigma; ratios require expected bin count ≥ 5 ; low-z satellite sample is small

Reference: unit Gaussian for satellite velocity relative to its host; sigma from the frozen effective radius; internal sampler test. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

08 Sky distribution, footprint and identity audit



L3870N4096/000: native fragmentation-group M_{PIN} [Msun/h], NOT M200m; native floor 10 particles; painting cut 32 particles = $2.931e12$ Msun/h.

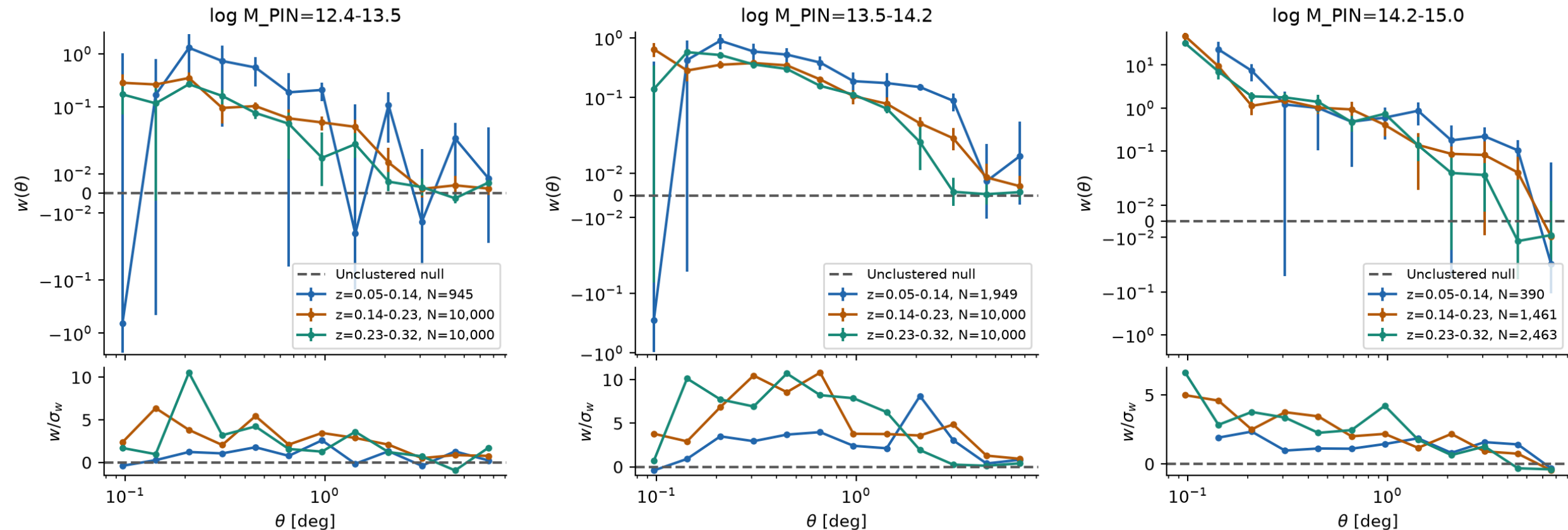
Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.

Selection: inner 65 deg; $M_r < -22$, $r < 20$; z_{cos} bins shown; $q = (1 - \cos \text{opening angle}) / (1 - \cos 65 \text{deg})$ | Sample: counts per row; 374 extra native group occurrences preserved; max native distance residual = $1.14e-04$

Uncertainty: conditional HOD 1 sigma; pull cells need variance ≥ 10 ; no cosmic-variance error (same fixed halo realization)

Reference: HOD at host centres, a model diagnostic; satellite boundary migrations are not integrated into the reference. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

09 Unfitted angular clustering: a diagnostic, not a calibration



L3870N4096/000: native fragmentation-group M_{PIN} [Msun/h], NOT M_{200m} ; native floor 10 particles; painting cut 32 particles = $2.931e12$ Msun/h.

Magnitudes: $0.1 M_r - 5 \log_{10}(h)$, AB; colour: rest-frame $0.1(g-r)$. Distances: comoving Mpc/h; peculiar velocities: proper km/s. Cone frame, not ICRS.

Selection: inner65 deg; $M_r < -22$, $r < 20$; z_{cos} and host-mass bins labelled; deterministic random subsampling | Sample: N per curve; $\leq 10,000$ data galaxies each; plotted null significance is diagnostic only

Uncertainty: 8 equal-area-region delete-one jackknife, 1 sigma; 3 randoms per data object; finite-random and covariance limitations apply

Reference: zero is an UNCLUSTERED NULL, NOT a predicted galaxy clustering model; no goodness-of-fit threshold. Completeness claims: $M_r \leq -22$, inner 65 deg, $0.05 \leq z_{\text{cos}} < 0.32$ only.

Checks, visible discrepancies and likely causes

Check	Result	Evidence / discrepancy	Interpretation / likely cause
Frozen inputs and model	PASS	Catalogue, input, calibration and model-source hashes verified before/after; no fitter called	Read-only audit
Independent LF target	PASS	9 complete bins; min N=246; max mock residual=3.026%; cumulative=0.00838%	Fixed target is fitted by the HOD; remaining mock deviations are stochastic and include boundary migrations
Native-mass completeness	LIMIT	10-particle native floor; 32-particle painting cut; max missing fraction in claimed bins=0.7532%	Fainter $M_r > -22$ and sparse bright bins are diagnostic only; not a physical halo-finder convergence test
HOD occupation	PASS	Conditional $\chi^2=88.9/61$ bins; $p=0.0114$, $\alpha=.01$	Mild excess scatter; $E[N] \geq 20$ and $\text{variance} \geq 10$ for Gaussian χ^2 . Compatible at the stated threshold, not a precision richness validation
Conditional colours	PASS	80 mixture/mean/width tests; 0 failures at familywise $\alpha=.01$	Same hodpy prescription tested, not independent observed colour data; red probability saturates in bright satellite bins
NFW radial / Gaussian velocity draws	PASS	12 mass/redshift-stratified KS tests; 0 failures; minimum $p=0.0242$	Model tests only; low-z satellites have limited statistics
Units, support and redshift closure	PASS	Integrity passed=True; Doppler closure= $2.95e-17$	R_{sat} is an effective model radius, not measured R_{200m}
Native PLC identities	CAVEAT	374 extra group-ID occurrences; 59 extra centrals across persistent IDs	Nearby distinct native crossing records retained; one central per unique PLC occurrence, not persistent ID
Sky / distance closure	CAVEAT	Native distance max relative mismatch= $1.14e-04$; sky residuals shown	Finite-precision native table/PLC; host-centre sky prediction omits satellite boundary migrations
Angular clustering	DIAGNOSTIC	Measured across 3 z and 3 host-mass bins, with spatial jackknife; no fitting	Nonzero clustering relative to the unclustered null is expected; no observed clustering or richness claim

No parameter was adjusted to improve any diagnostic. References: [hodpy](#), pinned revision d303bef8; [Smith et al. \(2017\)](#).